Numerical Analysis Assignment 4

刘金穗 - 2230018044

1

(a)

The matrix norm of matrix A is

$$||A||_F = 2 + \delta.$$

And the inverse of A is

$$A^{-1} = \frac{1}{\delta}\begin{bmatrix} -1 & 1 \\ \delta + 1 & -1 \end{bmatrix}$$

Its matrix norm is

$$||A^{-1}|| = \frac{1}{\delta}(2 + \delta)$$

Then, the condition number of A is

$$cond(A) = ||A|| \cdot ||A^{-1}|| = \delta + \frac{4}{\delta} + 4.$$

(b)

The exact solution for the system is $\mathbf{x}^* = (1, 1)^T$. The relative backward error is

$$\text{relative backward error} = \frac{||A\mathbf{x}_a - \mathbf{b}||_\infty}{||\mathbf{b}||_\infty} = \frac{\delta}{2 + \delta}.$$

The relative forward error is

$$\text{relative forward error} = \frac{||\mathbf{x}_a - \mathbf{x}||_\infty}{||\mathbf{x}||_\infty} = \delta + 2$$

Then

$$\text{error magnification factor} = \frac{\text{relative forward error}}{\text{relative backward error}} = \frac{(\delta + 2)^2}{\delta}.$$

2

The argument matrix of the system is

$$[A|\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 4 & 0 & 5 \\ 0 & 1 & 2 & 2 \\ 4 & 0 & 3 & 0 \end{array}\right].$$

Then,

$$[A|\mathbf{b}] \xrightarrow{r_1 \leftrightarrow r_3} \xrightarrow{r_2 \leftrightarrow r_3} \xrightarrow{r_1/2 \to r_1} \left[\begin{array}{ccc|c} 2 & 0 & 1.5 & 0 \\ 1 & 4 & 0 & 5 \\ 0 & 1 & 2 & 2 \end{array}\right]$$

is a strictly diagonally dominant system. Apply two step of Jacobi to it, the approximated solution is

$$\hat{\mathbf{x}}^{(2)} = (-0.7500, 1.2500, 0.3750)^T$$

Apply two step of Jacobi to it, the approximated solution is

$$\hat{\mathbf{x}}^{(2)} = (-0.2813, 1.3203, 0.3398)^T$$

3

(a)

$$\begin{aligned} x_i^{(k+1)} &= \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_i^{(k)} - \sum_{j=i+1}^{n} a_{ij} x_i^{(k+1)} \right) / a_{ii} + x_i^{(k)} - x_i^{(k)} \\ &= x_i^{(k)} + \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_i^{(k)} - \sum_{j=i}^{n} a_{ij} x_i^{(k+1)} \right) / a_{ii} \\ &= x_i^{(k)} + \frac{r_i^{(k+1)}}{a_{ii}}. \end{aligned}$$

(b)

$$\begin{aligned} x_i^{(k+1)} - x_i^* &= x_i^{(k)} - x_i^* + \frac{r_i^{(k+1)}}{a_{ii}}. \\ \varepsilon_i^{k+1} &= \varepsilon_i^k + \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_i^{(k)} - \sum_{j=i}^{n} a_{ij} x_i^{(k+1)} \right) / a_{ii} \\ \varepsilon_i^{k+1} &= \varepsilon_i^k + \left(-\sum_{j=1}^{i-1} a_{ij} \left(x_i^{(k)} - x_i^* \right) - \sum_{j=i}^{n} a_{ij} \left(x_i^{(k+1)} - x_i^* \right) \right) / a_{ii} \\ \varepsilon_i^{k+1} &= \varepsilon_i^k - \left(\sum_{j=1}^{i-1} a_{ij} \left(x_i^{(k)} - x_i^* \right) + \sum_{j=i}^{n} a_{ij} \left(x_i^{(k+1)} - x_i^* \right) \right) / a_{ii} \\ \varepsilon_i^{k+1} &= \varepsilon_i^k - \frac{R_i^{(k+1)}}{a_{ii}}. \end{aligned}$$